

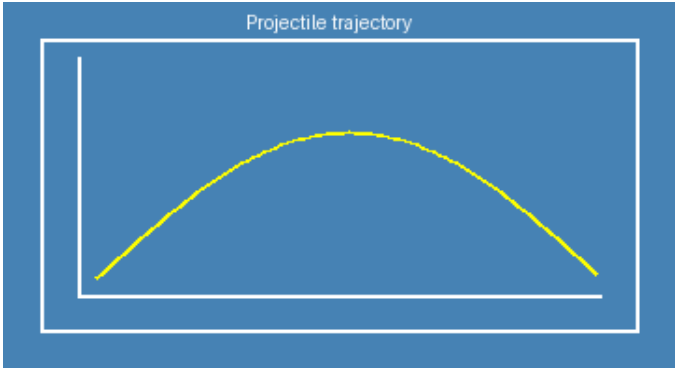
Chapter 1: Kinematics

In classical mechanics, the position of a particle is described as a function of time. The displacement, velocity, and acceleration form the foundation of kinematics. Consider a projectile launched at angle θ .

Key equations: $v(t) = v_0 + at$, $x(t) = x_0 + v_0t + \frac{1}{2}at^2$

Variable	Symbol	Unit	Value
Initial velocity	v_0	m/s	25.0
Acceleration	a	m/s ²	-9.81
Time	t	s	3.0
Displacement	x	m	45.9

Figure 1: Projectile trajectory



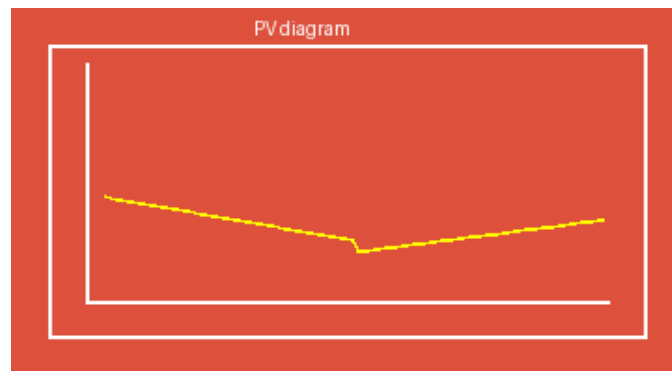
Chapter 2: Thermodynamics

The laws of thermodynamics govern the transfer of energy in physical systems. Entropy is a measure of disorder, and the second law states that the total entropy of an isolated system never decreases.

Key equations: $\Delta S = Q/T$, $PV = nRT$, $dU = \delta Q - \delta W$

Property	Symbol	Unit	Example
Temperature	T	K	300
Pressure	P	Pa	101325
Volume	V	m ³	0.0224
Entropy	S	J/K	154.8

Figure 2: PV diagram



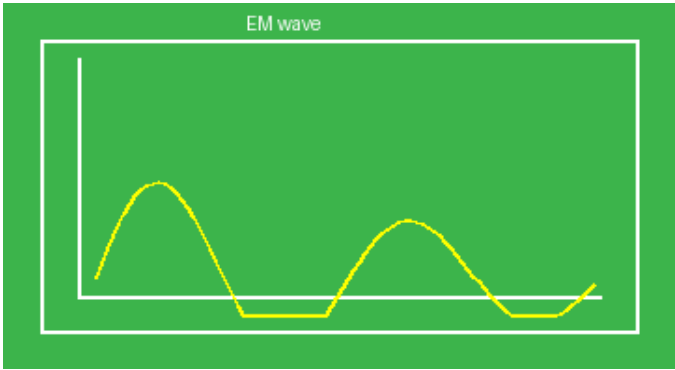
Chapter 3: Electromagnetism

Maxwell's equations unify electricity and magnetism into a single framework. The curl of the electric field relates to the time derivative of the magnetic field, producing electromagnetic waves.

Key equations: $\nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t$, $\nabla \cdot \mathbf{B} = 0$, $\mathbf{F} = q\mathbf{E} + q\mathbf{v} \times \mathbf{B}$

Quantity	Symbol	SI Unit	Typical Value
Electric field	E	V/m	3.0×10^6
Magnetic field	B	T	1.5
Charge	q	C	1.6×10^{-19}
Permittivity	ϵ_0	F/m	8.85×10^{-12}

Figure 3: EM wave



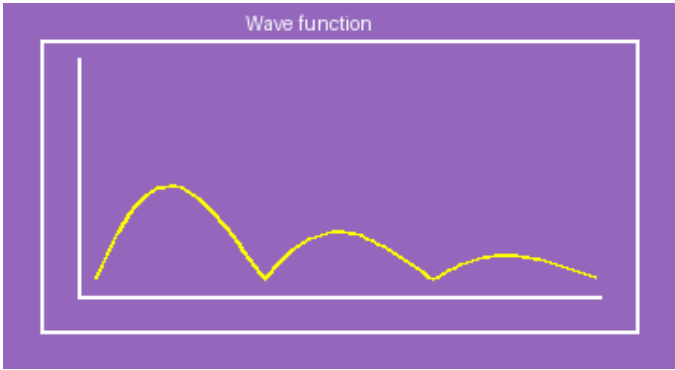
Chapter 4: Quantum Mechanics

At the atomic scale, particles exhibit wave-particle duality. The Schrödinger equation describes how the quantum state of a system evolves in time. Measurement collapses the wave function.

Key equations: $i\hbar \frac{\partial \psi}{\partial t} = \hat{H}\psi$, $\Delta x \Delta p \geq \hbar/2$, $E = \hbar \omega$

Constant	Symbol	Value	Unit
Planck's constant	$\hbar$	1.055×10^{-34}	J·s
Electron mass	m_e	9.109×10^{-31}	kg
Bohr radius	a_0	5.292×10^{-11}	m
Fine structure	α	1/137.036	—

Figure 4: Wave function



Chapter 5: Statistical Mechanics

Statistical mechanics bridges microscopic particle behavior and macroscopic thermodynamic properties. The partition function encodes all equilibrium information of a canonical ensemble.

Key equations: $Z = \sum \exp(-E_i/kT)$, $F = -kT \ln Z$, $\langle E \rangle = -\partial \ln Z / \partial \beta$

Ensemble	Fixed	Fluctuates	Potential
Microcanonical	E, V, N	—	$S(E,V,N)$
Canonical	T, V, N	E	$F(T,V,N)$
Grand canonical	T, V, μ	E, N	$\Phi(T,V,\mu)$
Isothermal-isobaric	T, P, N	E, V	$G(T,P,N)$

Figure 5: Boltzmann distribution

